Outburst activity of the symbiotic system AG Dra

L. Hric, R. Gális, L. Leedjärv, M. Burmeister, E. Kundra

Monthly Notices of the ROYAL ASTRONOMICAL SOCIETY

MNRAS **443,** 1103–1112 (2014) doi:10.1093/mnras/stu1162

Outburst activity of the symbiotic system AG Dra

L. Hric,[1]★ R. Gális,[2] L. Leedjärv,[3] M. Burmeister[3] and E. Kundra[1]

[1]*Astronomical Institute, Slovak Academy of Sciences, Tatranská Lomnica 059 60, Slovak Republic*
[2]*Faculty of Science, P. J. Šafárik University, Park Angelinum 9, Košice 040 01, Slovak Republic*
[3]*Tartu Observatory, 61602 Tõravere, Estonia*

Accepted 2014 June 9. Received 2014 June 9; in original form 2014 March 25

ABSTRACT

AG Dra is a well-known bright symbiotic binary with a white dwarf and a pulsating red giant. Long-term photometry monitoring and a new behaviour of the system are presented. A detailed period analysis of photometry as well as spectroscopy was carried out. In the system of AG Dra, two periods of variability are detected. The longer one around 550 d is related to the orbital motion and the shorter one around 355 d was interpreted as pulsations of the red giant in our previous article. In addition, the active stages change distinctively, but the outbursts are repeated with periods from 359–375 d.

Key words: binaries: symbiotic – stars: individual: AG Dra – stars: oscillations.

1 INTRODUCTION

AG Dra is a classical symbiotic binary, type S. It is one of the best-studied symbiotic systems, thanks to its relatively high brightness (about 9.8 mag in quiescence and about 8.3 mag in the largest outbursts in *V*), high Galactic latitude ($b = 41^\circ$) favourable for observations and low extinction ($E(B-V) = 0.05$: Mikolajewska et al. 1995).

The cool component of AG Dra is of a relatively early spectral type, with classification in the range K0–K4 and low metallicity, [Fe/H] = −1.3 (Smith et al. 1996). The mass of the giant was estimated as 1.5 $M_\odot$ by Kenyon & Fernandez-Castro (1987). Its luminosity could be higher than that of standard class III (Huang, Friedjung & Zhou 1994; Mikolajewska et al. 1995). Smith et al. (1996) found an overabundance of elements heavier than Fe (mainly Ba and Sr) and thus classified the giant as a barium star. Such stars are on average more luminous than standard giants of the same spectral class and they could be more capable of invoking symbiotic activity in the binary system, due to their higher mass loss rate. The radius of the giant was estimated to be $\sim 35\,R_\odot$ by Zamanov et al. (2007) and Garcia (1986) found an orbital separation of 400 $R_\odot$. If we estimate the Roche-lobe radius on the basis of the above-mentioned values, we can conclude that the giant probably does not fill its Roche lobe.

The hot component of AG Dra is considered to be a white dwarf sustaining a high luminosity ($\sim 10^3\,L_\odot$) and temperature ($\sim 10^5$ K) due to the thermonuclear burning of accreted matter on its surface (Mikolajewska et al. 1995). The accretion most likely takes place from the stellar wind of the cool giant. Both components are in a circumbinary nebula, partially ionized by the hot component.

AG Dra undergoes characteristic symbiotic activity with alternating quiescent and active stages. Active ones consist of several outbursts repeating at about a one-year interval. The amplitudes of the outbursts decrease towards longer wavelengths, from ∼1 mag in *V* to ∼3 mag in *U*. Periodical outbursts and their relation to the orbital motion of the binary system have been a matter of long-term debate. While there is general agreement that the orbital period of AG Dra is about 550 d (Meinunger et al. 1979; Gális et al. 1999; Fekel et al. 2000), there are variations on shorter time-scales (350–380 d) presented by Bastian (1998), Friedjung et al. (1998, 2003) and some others. Understanding the nature and mechanism of this variability is crucial in order to explain the outburst activity of AG Dra and other classical symbiotic stars.

González-Riestra et al. (1999) showed on the basis of the analysis of all *International Ultraviolet Explorer* (*IUE*) observations that there are two types of outburst: cool and hot. During a cool outburst (e.g. 1981–83 and 1994–96), the hot compact object has a temperature ≈90 000 K, thus lower than that in the quiescent stage. Hot outbursts (e.g. 1985–1986) are characterized by temperatures above ≈130 000 K and considerably lower optical and UV brightness.

Skopal et al. (2009) studied the supersoft X-ray versus optical/UV flux anticorrelation of AG Dra. They concluded that such behaviour is caused by the variable wind from the hot component of the system. Shore, Genovali & Wahlgren (2010) found that variations in the Raman features ratio during the outburst are consistent with the disappearance of the O VI far-UV resonance wind lines from the white dwarf. The observations support the suggestion that the soft X-ray and UV variations are due to the expansion of the envelope and decreasing effective temperature of the accreting star. The outburst activity (2006–2008) of AG Dra was studied by Munari et al. (2009) using photometric and spectroscopic observational material. The first outburst of this activity stage was of the cool type and was followed by a fainter maximum of the hot type after 375 d. Contini & Angeloni (2011) mentioned that the collision of the wind from

★ E-mail: hric@ta3.sk

Table 1. List of observational nights at particular observatories: SP – Skalnaté Pleso; SL-G1 – Stará Lesná, pavilion G1; SL-G2 – Stará Lesná, pavilion G2; VM – Valašské Meziříčí.

Observatory	N_{nights}	Period	Period [MJD]
SP + SL-G2	123	18.5.1999–16.3.2012	51 317–56 003
SL-G1	31	8.4.2006–27.9.2009	53 833–55 102
VM	151	24.4.1997–24.4.2005	50 563–53 485

the white dwarf with the dusty shells ejected from the red giant leads to fluctuations in the *U* band during the major outburst. The long-term spectroscopic study of AG Dra was presented by Shore et al. (2010). They discussed the effects of environment and orbital modulation in this system. Recently Leedjärv & Burmeister (2012) demonstrated the spectroscopic behaviour of AG Dra before, during and after the last outburst.

Recently Formiggini & Leibowitz (2012) accomplished complex period analysis of the historical optical light curve (LC) of AG Dra, covering the last 120 years. They found that the period of the outbursts was 373.5 d and identified a cyclical behaviour with a quasi-period of 5300 d. The last is the time interval between the start of the active stages induced by the solar-like cycles of magnetic activity. The combined effect of the 5300- and 373.5-d cycles augments the postulated cool giant pulsations of the red giant with a period of 350 d (Formiggini & Leibowitz 2012). However, the proposed rotational period of the giant of 1160 d in their model, and in particular its retrograde rotation, seems to be somewhat artificial and not well justified physically.

In the present article, we re-analyse the *UBVR* LCs of AG Dra and also apply time-series analysis to the spectroscopic data. We use all the available photometry, starting from the compilation of photographic observations by Robinson (1969) and proceeding with a large amount of photoelectric observations (see hereafter). Variability of the emission lines provides additional information on the physical state of the AG Dra system.

2 PHOTOMETRIC AND SPECTROSCOPIC DATA

The new photoelectric and CCD observational material[1] was obtained at the observatories at Skalnaté Pleso (SP), Stará Lesná (SL-G1 and SL-G2) and Valašské Meziříčí (VM). At observatories SP and SL-G2, identical Cassegrain telescopes with a diameter of 0.6 m were utilized. One-channel photoelectric photometers with digital converters were used, as well as standard *UBVR* Johnson filters. CCD photometry was performed at SL-G1 using the 0.5-m telescope. The SBIG ST10 MXE CCD camera with a chip of 2184 × 1472 pixels and the $UBV(RI)_{\mathrm{C}}$ Johnson–Cousins filter set were mounted at the Newtonian focus.

A Schmidt–Cassegrain telescope (280/1765 mm) equipped with CCD ST-7 and a set of VR filters was used at VM observatory. From JD 245 0563 to JD 245 1179, photographic films were used as a detector. The number of observational nights, observational intervals in date format as well as in Julian days for particular observatories, are listed in Table 1. In addition, we used the same data that had already been analysed and discussed in our previous article (Gális et al. 1999) and published photoelectric *UBV* photometry by Belyakina (1965, 1969), Skopal et al. (2002, 2004, 2012), Leedjärv et al. (2004) and Munari et al. (2009), as well as photographic measurements by Robinson (1969) and Luthardt (1983).

To support the model of cool giant pulsations, we obtained the photometric observations as instrumental ΔR_{i} magnitudes. These magnitudes were not transformed to the international system, due to the lack of comparison star *R* magnitudes. These observations were secured in the period from JD 244 9557 (1994 July 23) to JD 245 5102 (2009 September 27) at the SP and SL-G1 observatories and some of them were published by Skopal & Chochol (1994), Skopal et al. (1995, 2002, 2004, 2012), Hric et al. (1996) and Skopal (1998).

Intermediate-dispersion spectroscopy of AG Dra was carried out at the Tartu Observatory in Estonia. The data used in this article covered the interval from JD 245 0702 (1997 September 11) to JD 245 5651 (2011 March 31). Altogether, 211 spectra of AG Dra were used for the period analysis. The spectroscopic material was taken by the 1.5-m telescope equipped with a Cassegrain grating spectrograph. The equipment and method of observations were described in Leedjärv et al. (2004) and Leedjärv & Burmeister (2012). The majority of the spectra were recorded in two spectral regions, which we call red and blue. Red spectra were taken with a dispersion of 0.47 Å pixel^{-1} in Hα and include emission lines of Hα, He I 6678 Å and the Raman-scattered O VI line at 6825 Å. The blue spectra included at least emission lines of Hβ, He II 4686 Å and He I 4713 Å. The dispersion of the blue spectra was about 0.57 Å pixel^{-1} in Hβ. The spectra were reduced using the software package MIDAS provided by the European Southern Observatory (ESO). The wavelength-calibrated spectra were normalized to the continuum and positions, peak intensities and equivalent widths (EW) of the emission lines were measured.[2]

3 MORPHOLOGY OF THE LIGHT CURVES AND PERIOD ANALYSIS

In this section, we describe the results of re-analysis of the historical LC from the photographic data up to 1966. We have also presented the results of the analysis of *UBVR* photometry from 1974 to the present day. The results of principal component analysis of *UBV* photometry are also discussed.

Period analysis of the observational data was performed using an advanced implementation of the Date-Compensated Discrete Fourier Transform. We used a Fisher Randomization Test (Monte Carlo Permutation Procedure) to calculate two complimentary False Alarm Probabilities (FAPs) for determining the significance of a given period *P*: FAP 1 represents the proportion of permutations containing a period with a peak/valley higher (respectively lower) than the peak/valley of *P* at any frequency. It is the probability that there is no period in the period window (power spectrum) with value *P*. FAP 2 represents the proportion of permutations containing a period with a peak/valley higher (respectively lower) than the peak/valley of *P* at exactly the frequency of *P*. It is the probability that the observation data contain a period that is different from *P*. This test executes the selected period analysis calculation repeatedly (at least 100 times), each time shuffling the magnitude values of the observations into the form of a new randomized observation set, but keeping the observation times fixed (Press et al. 1992). FAPs with value below 0.01 (1 per cent) mostly indicate significant periods. FAPs above 0.20 (20 per cent) mostly relate to artefacts in the data, instead of true periods. *In this article, we consider a period*

[1] The photometric data are available upon request from the authors.

[2] The spectroscopic data are available upon request from the authors.

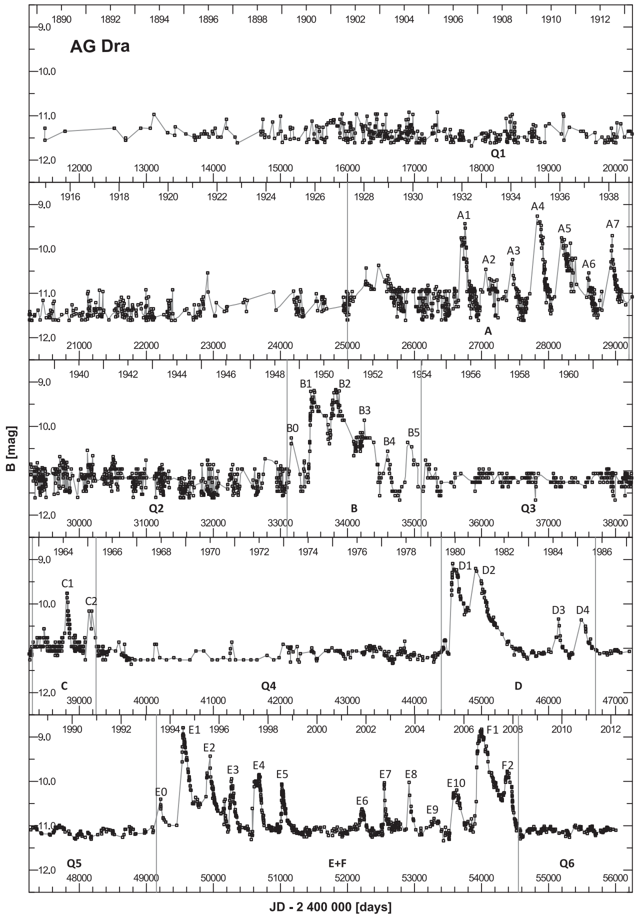

Figure 1. The historical LC of AG Dra over the period 1889–2012, constructed on the basis of photographic and *B* photoelectric observations. The data sources used are quoted in the text. The LC is divided into active (A–F) and quiescence (Q1–Q6) stages by vertical lines. The thin curve shows a spline fit to the data points.

significant if the FAPs for the period are less than 0.01, although larger values up to 0.2 may occasionally be allowed.

In addition, we created a spectral window to confirm that the periods found in the previous steps are not artefacts of the observing rate. We performed detailed period analysis for some complicated cases (e.g. during active stages), in which the response of the examined period *P* was removed from the data and the power spectra of such residuals were studied with a focus on the persistent presence of aliases and harmonics of this period. The minimum period error (or period uncertainty) of period *P* was determined by calculating a 1σ confidence interval on *P*, using the method described by Schwarzenberg-Czerny (1991).

3.1 The light curve between the years 1890 and 1966

The first historical photometric observations were dated to the end of the 19th century (Fig. 1). During the period (1890–1996), the AG Dra system underwent three phases of activity: the first one between the years 1932 and 1939, the second one between 1949 and 1955 and the third one between 1963 and 1966. In total, we recognized 15 outbursts in this period: seven during the first active phase, six during the second one and two outbursts during the last one.

We present the results of the re-analysis of this data with the contemporary method described in Section 3. We divided the LC into quiescent and active stages (marked by vertical lines in Fig. 1) and carried out a period analysis independently for each stage. We present the results of this analysis for the given stages in Table 2, as well as in Fig. 2. We conclude that the historical LC shows both known periods: ≈550 d (orbital period) and ≈350 d (the period of the postulated pulsation of the red giant: Gális et al. 1999). Besides

Table 2. Results of period analysis of the historical LC of AG Dra, constructed on the basis of photographic and *B* photoelectric observations. T_{start} marks the beginning and T_{end} the end of the given stage. During stages Q1, Q2 and Q3, the system was quiescent, whereas stages A, B and C are active.

Phase	T_{start} [MJD]	T_{end} [MJD]	Significant periods [days]
Q1	11 500	25 000	551.0 ± 3.5; 348.8 ± 2.1
A	25 500	29 200	371.3 ± 4.6; 548.0 ± 8.1
Q2	29 200	33 100	399.6 ± 5.6
B	33 100	35 100	352.7 ± 5.9
Q3	35 100	38 300	438.0 ± 10.7; 534.0 ± 23.9
C	38 300	39 250	380.2 ± 10.2
Q4	39 250	44 400	550.0 ± 10.3; 350.9 ± 4.8

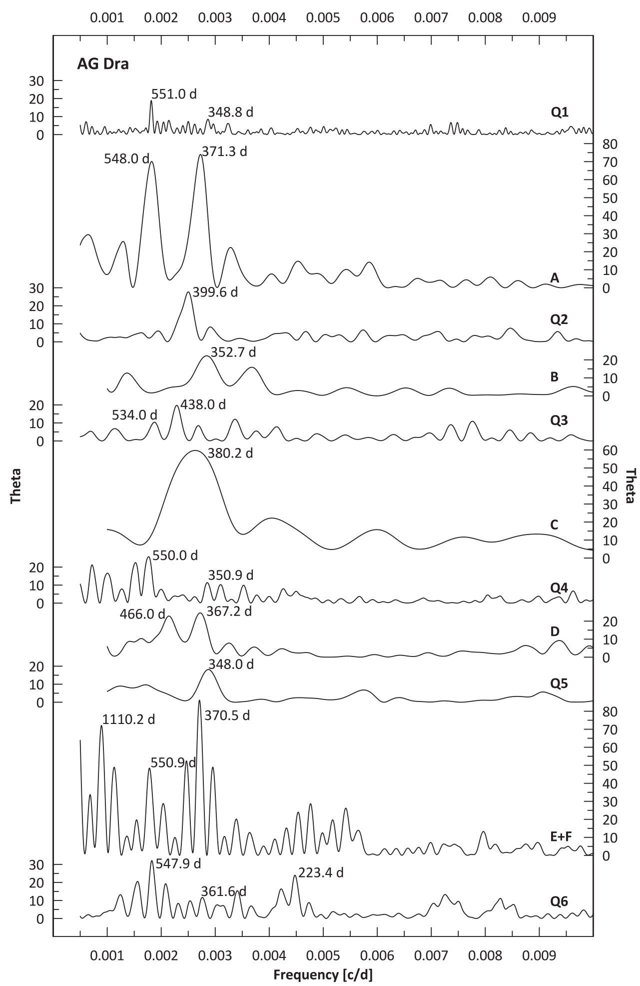

Figure 2. Power spectra of AG Dra taken from historical photographic as well as photoelectric and CCD data in the *B* filter for particular stages of quiescence (Q1–Q6) and activity (A–F). Significant periods are marked with their values.

these periods, the analysis gave us the period 370–380 d, which is present in the active stages A and C. This period is related to the recurrence of the individual outbursts that occurred in these active stages. The periods around 400 d (Q2) and 440 d (Q3) are probably not real and are caused by the low amplitude of light variations in comparison with the scatter of these photographic data.

Between the years 1889 and 1927 (stage Q1), AG Dra was in quiescence, with a mean photographic brightness of 11.02 mag. Small variations of the LC revealed the presence of both known periods. After this, at least 38 years of quiescence, the system went into an active stage (assigned as A in Fig. 1). Up until the year 1938, we can recognize seven major outbursts occurring approximately every 300–400 d. The individual outbursts have a rather steep rise to maximum and a fast decline. The outburst in the year 1932 (A1) had two maxima, with a second (more prominent) maximum following the first one after ≈50 d. The system reached the maximal brightness of 8.9 mag during the fourth outburst (A4). For the next almost 11 years, a quiescent stage followed with only small semi-regular variations. The period analysis gave us a period with a value ≈ 400 d, which is more probably related to the data distribution in this part of the LC (see stage Q2 in Fig. 1) than real brightness variability of the AG Dra system. This distribution might have drowned the expected orbital or probable pulsation period.

An epoch of intense activity was observed between the years 1949 and 1955 (stage B). The overall morphology of the LC during this active stage is rather different from that of stage A. The peaks have an analogously steep rise to maximum, with the highest achieved brightness 8.8 mag. The decrease is considerably slower and, unlike stage A, the system spends most of this period above 10.5 mag. The outbursts follow one after another, ≈350 d. The second and third outbursts (B2 and B3) might again have secondary maxima approximately 50 d after the primary ones.

Stage Q3 (1955–1963) is characterized by semi-regular featureless brightness variations with period ≈440 d. This behaviour probably influenced the value of the orbital period (534 d) obtained by our analysis. The system became active after the year 1963, which resulted in the shortest (1963–1966) and least active (only 2–3 outbursts) stage during the whole observed LC of AG Dra. The period analysis of this active stage C revealed the presence of a 380-d period.

3.2 The light curve after 1966

For complete coverage of the 124-year history of AG Dra, we are missing an eight-year section on the LC between JD 243 9355 (1966 August 17) and JD 244 2109 (1974 March 2). In the archive of the Sonnenberg Observatory, we have found two articles (Splittgerber 1974; Luthardt 1983) where the LC behaviour during this period is described, but the data were not available. We extracted the photometric data from the LC depicted in fig. 2 of the article by Luthardt (1983) to cover the missing part of the historical LC. It was obvious from the completed LC (Fig. 1) that after outburst C the system fell into a long-term quiescent stage (assigned as Q4). The activity of the system was very low during this period and the average photographic brightness was 10.9 mag. The period analysis of this part of the LC shows unambiguously the presence of the 550-d orbital as well as the 350-d postulated pulsation period.

The quiescent stage Q4 is also partly covered by photoelectric observations. The first photoelectric *UBV* photometry of AG Dra was obtained by Belyakina from 1962–1967 (Belyakina 1965, 1969). Since 1974, the system of AG Dra has been observed systematically, mainly photoelectrically in *UBV*. The merit of good coverage

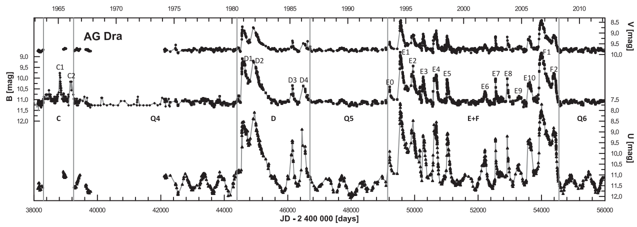

Figure 3. *UBV* LCs from the period 1963–2012 with marked active stages (C, D, E and F) and quiescent ones (Q4, Q5 and Q6). Particular outbursts are assigned as C1–C2, D1–D5, E0–E10 and F1, F2. The thin curves show spline fits to the data points.

of the LC has been our campaign (Hric & Skopal 1989), also. The benefits of these data were the improvement of the orbital period (≈550 d) and the discovery of the shorter period (≈350 d), which was interpreted as the pulsation of the cool component (Gális et al. 1999). The photoelectric LCs of AG Dra are depicted in particular filters in Fig. 3. It is obvious from the LCs in *U*, *B* and *V* bands that the amplitudes of variations decrease towards longer wavelengths. During the activity of the system, the LCs are in strong correlation in all bands. The variations in quiescence are similar in *B* and *V*, but differ in the *U* bands.

In 1980, the system entered the active stage D, where outbursts were characterized by a fast increase and slower decrease as in stage B. The individual outbursts are clearly recognized in Fig. 3. It is worth noting that the active stage D was interrupted by the interval (1983–1985), when the system manifested the behaviour typical for quiescent stages. AG Dra was in a quiescent stage (Q5) for the next seven years. In 1993, new activity (E) started with a small outburst followed by the next five prominent outbursts consecutively at ≈350–390 d intervals. Activity of the system was again interrupted by a short quiescent-like interval in 1999–2001 (similar to the behaviour that happened during the active stage D) and continued by the next five outbursts, which finished at the beginning of 2006. Such long-term activity has never been observed during more than one century of photometric history of AG Dra. This activity was immediately (without a quiescence phase) followed by the large outburst (F1) in 2006 when the brightness in *U* reached 8 mag. This new phase of activity (F) had two (up to JD 245 5000) outbursts, followed by a fast drop to the deep brightness minimum (11.8 mag in the *U* band). Since 2008, the symbiotic system AG Dra has been in a quiescent stage (Q6).

From the LC in Fig. 3, we can distinguish two kinds of outbursts. There are the main double-peaked outbursts (B1 and B2, D1 and D2, E1 and E2, F1 and F2) with a very rapid increase of brightness, small drop of magnitude between peaks and slow decrease to quiescence level. The identification of double-peaked outbursts is ambiguous during active stage A and such a feature is totally missing in the weak active stage C. This type of outburst defines the beginning of each new cycle of activity of AG Dra, with time interval 12–16 years. Moreover, there are outbursts with a sharper shape and lower amplitude, which are observed after the main outburst and sometimes during the whole cycle of activity (B3–B5, C1–C2, D3–D4, E3–E10).

Our statistical analysis of the photoelectric and CCD observations shows that the LCs in *U*, *B* and *V* filters were very well correlated (correlation coefficients ≈0.9) during the active stages (D, E and F in Fig. 3). During the quiescent stages (Q4–Q6 in Fig. 3), the correlation coefficients of the LCs in bands *U* and *B* as well as one of the LCs in bands *U* and *V* are less than 0.5. On the other hand, the variations in the *B* and *V* bands are correlated quite well during

Table 3. The results of period analysis of particular stages between 1963 and 2012. LCs for each filter were analysed separately. T_{start} is the beginning and T_{end} the end of the given stage. The periods are in order according to their significance.

Phase	T_{start} [MJD]	T_{end} [MJD]	Significant periods [days] *U* band	*B* band	*V* band
Q4	39 250	44 400	551.0 ± 2.4	550.0 ± 10.3; 350.9 ± 4.8	349.6 ± 13.9; 550.0 ± 49.7
D	44 400	46 700	371.9 ± 5.5	367.2 ± 8.1; 466.0 ± 15.2	372.5 ± 6.1
Q5	46 700	49 150	553.6 ± 4.0	348.0 ± 6.7	350.1 ± 7.3
E + F	49 150	54 550	371.2 ± 1.8	370.5 ± 1.9; 1 110.2 ± 18.5[a]; 550.9 ± 9.8	370.5 ± 1.8
Q6	54 550	continue	549.3 ± 2.7	547.9 ± 6.4; 223.4 ± 5.3[b]; 361.6 ± 5.3	357.3 ± 19.8

Notes: [a]The period of 1110.2 ± 18.5 d is probably only the double of 550.9 ± 9.8 d. [b]The period of 223.4 ± 5.3 d is the one-year alias of 547.9 ± 6.4 d. The global morphology of active stages D and E+F is possibly very well described by sinusoidal variations with periods around 1110 d (or double 2220 d) and 2500 d (or double 5000 d).

the quiescence. This result showed that brightness variations during the quiescent stages of AG Dra in the various bands were caused by different physical mechanisms.

We performed period analysis of LCs in individual bands (*U*, *B* and *V*) with the same methods used for the period 1890–1966. The results of this analysis are presented in Table 3. The power spectra of AG Dra in *U*, *B* and *V* filters for active stages as well as for quiescence are depicted in Fig. 4. As we have shown by our period analysis of the quiescent stages, the LC in the *U* band is clearly dominated by variations with orbital period ≈550 d. In the *B* and *V* bands, we found a shorter period (≈350 d); however, its value in each quiescent stage changed slightly. Period analysis of the active stages revealed many significant periods. Our detailed analysis showed that most of the periods were more likely related to the complex morphology of the LCs during the active stages than the real variability present in this symbiotic system.

The significant period with a value of around 370 d is related to the distribution of individual outbursts, which dominate the LC in the active stages. It should be noted, however, that the value of this period varies with wavelength and is different for the individual active stages. Statistical analysis of the outburst distribution shows that the median of the time interval between the individual outbursts is 365 d, while the time intervals vary from 300 to 400 d without an apparent long-term trend.

We can conclude that the results of period analysis of *UBV* LCs strongly confirm both known periods. The dominance of the orbital period in the *U* filter is in agreement with the model by Gális et al. (1999).

The stronger influence of the second period at longer wavelengths suggests it is related to the behaviour of the red giant and is probably caused by its pulsations, as we proposed in the article mentioned above.

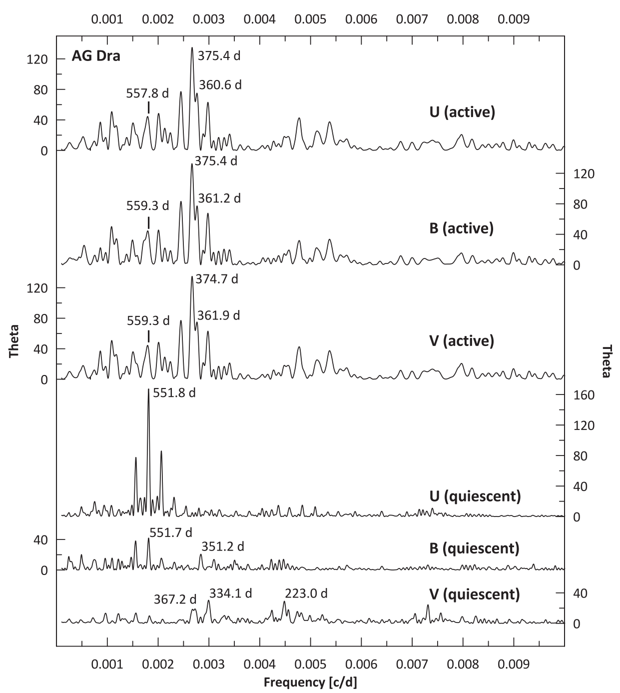

Figure 4. Power spectra of AG Dra taken from photoelectric and CCD data in *U*, *B* and *V* filters for active (D, E+F) and quiescent (Q4–Q6) stages. Power spectra for active stages in *U*, *B* and *V* filters were obtained after removing long-term periods around 1500 and 5400 d, which are related to the global morphology of these active stages.

3.3 Differential R_i photometry of AG Dra during 1994–2009

As we have shown in the previous section, two periods are present in the LCs of AG Dra. For the 350-d period, Gális et al. (1999) proposed interpretation as pulsations of the cool giant. To support this model, we carried out a period analysis of photometric observations in instrumental ΔR_i magnitude.

We had at our disposal 226 nights of observations covering part of the active phase (E in Fig. 3). All observations are depicted in Fig. 5. Pronounced outbursts, as well as variations during quiescence, are modulated with the period $P_R = (371.0 \pm 2.2)$ d. The presence of this period at longer wavelengths during quiescence of AG Dra probably suggests its relation to the cool component in this system. The red giant pulsations proposed by Gális et al. (1999) are a possible source of modulations with this period.

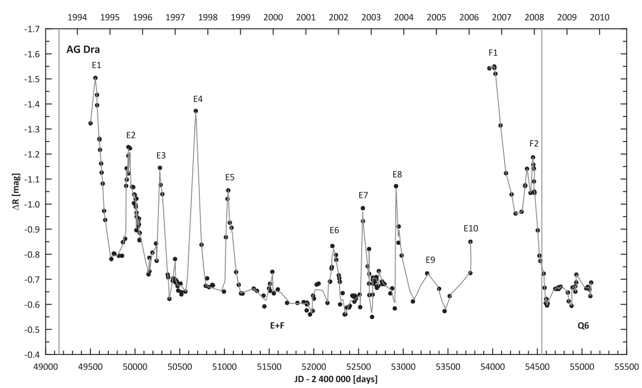

Figure 5. Light curve of AG Dra in instrumental ΔR_i mag. The thin line shows a spline fit to the data points.

3.4 Results of principal-component analysis of *UBV* photometry

We accomplished the principal-component analysis (PCA) of *UBV* photometry from the period 1974–2006. We analysed the whole LC and then particular stages (Q4, D, Q5 and E) separately. PCA is a type of linear orthogonal transformation (also known as the discrete Karhunen–Loève transform or Hotelling transform) used to convert a set of observations of possibly correlated variables into a set of values of linearly uncorrelated variables, called principal components. This transformation is defined in such a way that the first principal component (PC1) has the largest possible variance and each succeeding component in turn has the highest variance possible under the constraint that this component is orthogonal to the preceding ones.

For this analysis, only those observations with magnitudes in all three filters were used. We composed the input values of three vectors $\boldsymbol{U} \equiv (U_{t1}, U_{t2}, \ldots, U_{tN})$, $\boldsymbol{B} \equiv (B_{t1}, B_{t2}, \ldots, B_{tN})$ and $\boldsymbol{V} \equiv (V_{t1}, V_{t2}, \ldots, V_{tN})$, where U_{ti}, B_{ti} and V_{ti} were magnitudes in particular filters in time t_i and N is the number of observations. The output of the method is a ternary of vectors PC1, PC2 and PC3, which represents the principal components of the input observations. They have N dimensions like the input vectors and therefore it is possible to assign time t_i and create the LCs of principal components (see Fig. 6). The diagrams in this figure represent the LCs of principal components computed for the period 1974–2006. If we compare the components with the original LCs, we can see that the first component PC1 includes most of the significant variations in the original data. The second component PC2 contains the variation related to the orbital motion. The last component PC3 does not contain significant information about the binary system, only the noise caused by errors in the data.

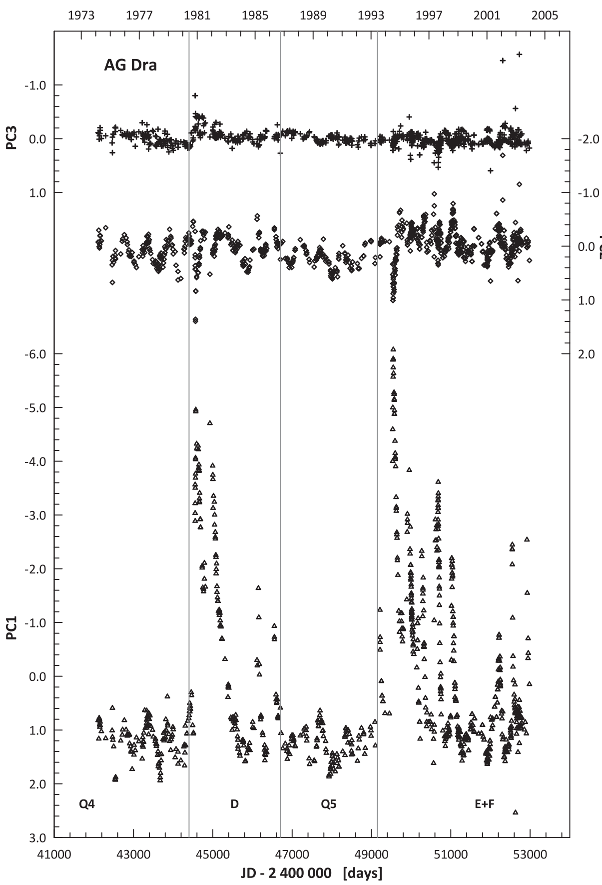

Figure 6. Principal components PC1, PC2 and PC3 for AG Dra *UBV* LCs.

Table 4. Variation analysis of principal components over the whole period (1974–2006).

Component	Variation	Cumulative sum
PC1	2.889	2.889
PC2	0.089	2.979
PC3	0.021	3.000
Original LCs:	3.000	

It is possible to quantify the informational content in particular components by variation analysis. The amplitudes of variations in the U, B and V bands decrease towards longer wavelengths. We scaled the original U, B and V LCs to unit variations to secure the same weights for the LCs being input into the analysis. It is evident from Table 4 that output components are not equivalent. The variation of PC1 makes up as much as 96.3 per cent of the variation in the input data and so represents 96.3 per cent of the entire information content. That is caused by the mechanism of PCA, which selects the principal component PC1 so that its correlation with the input signal is the highest. The total variation of principal components is equal to the total variation of the input data. It means that the principal components contain all information about the input LCs.

In the next step, the LCs were divided into particular stages (Q4, D, Q5 and E) and variation analysis was accomplished for each stage separately. This analysis gave similar results to the case of the total LCs.

Apart from variation analysis of particular components, it is interesting to study their correlations as well as the correlations with the original LCs. We analysed correlations among all combinations of LCs (U, B, V) and principal components (PC1, PC2, PC3). The correlations of the LCs in particular filters are very interesting. During activity, these correlation coefficients are very high ($\geq$0.97). The behaviour of U, B and V LCs is equivalent, because outbursts are dominating features in all bands. In this manner, we detected high correlation with PC1, too. The behaviour of quiescent stages is different. The LCs in the U and B bands have a correlation coefficient $\approx$0.7, but those in the U and V bands have a correlation coefficient only $\leq$0.5, because the orbital period dominates in the U band while the probable pulsation period is more significant in the V band.

The brightness variations related to the orbital motion as well as the probable red giant pulsation should be present in all three bands, but with different amplitudes. The method of PCA transforms the variations into the first principal component. Moreover, the advantage of PC1 period analysis in comparison with analysis in particular photometric bands is that the first principal components contain most of the information, the noise is suppressed (it is isolated in the last component) and the detected periods have higher statistical significance. Therefore we accomplished the period analysis of PC1. The results of this analysis for particular stages are listed in Table 5. The principal component PC1 contains two significant periods around 550 and 350 d; the individual values of the detected periods vary from 343–368 d and from 526–576 d for the postulated pulsation and orbital periods, respectively. The presence of a 368-d period could possibly be explained by the time distribution of individual outbursts during active stages. The large scatter

Table 5. The results of period analysis of principal component (PC1) for particular stages in 1974–2006.

ID	Significant periods [days]
Q4	563.9 ± 11.2; 343.0 ± 5.6
D	484.5 ± 7.2; 367.9 ± 4.1
Q5	575.6 ± 11.5; 348.1 ± 3.8
E	367.9 ± 1.1; 526.1 ± 3.6

in values of the orbital period (larger than the typical error of period determination) as well as the presence of a 485-d period is unclear.

Fig. 6 suggests that the principal component PC2 contains only the orbital variations. The period analysis of PC2 over the whole time interval (1974–2006) revealed the orbital period $P_{\rm PC2} = (555.9 \pm 1.2)$ d. This orbital period also dominates in the period analysis of particular stages (Q4, D, Q5 and E), except for D, where the detected period is ≈480 d. The period analysis of PC2 also confirmed the presence of the shorter period ≈340–370 d in all particular stages, but with lower significance than in PC1. In the LC of PC2 (see Fig. 6), faint suggestions of outbursts in stages D and E are also visible. The period analysis of PC3 did not give the presence of any significant period.

The activity of the system is also manifested in differential $R_{\rm i}$ photometry. From all observational nights, data with magnitudes in all four filters were selected. For the 98 observations, variation analysis as well as the period analysis of principal components was performed. The variation analysis showed that, during the active stages, the LC in the ΔR_i band correlated well with LCs in all other bands. For correlation coefficients, the following values were obtained: $C_{\rm RU} = 0.95$, $C_{\rm RB} = 0.97$, $C_{\rm RV} = 0.97$. This means that outbursts also dominate in the red LCs. Period analysis of the principal component PC1 disclosed the significant presence of the 'postulated' pulsation period $P_{\rm pul} = (352.4 \pm 6.3)$ d. The next significant period, $P_{\rm orb} = (544.2 \pm 17.4)$ d, is close to the value determined for the orbital period. In the LCs for PC2 and PC3, only unreal periods around 700 d were found. The last component PC4 did not reveal any significant periods.

4 PERIOD ANALYSIS OF SPECTROSCOPIC DATA

For the period analysis of the radial velocities, we used the method of Fourier harmonic analysis (Andronov 1994), which fits the first harmonic term of a trigonometric polynomial approximation, and the Variable Stars Calculator (Breus 2003), which utilizes the Lafler–Kinman–Kholopov method. For verification of the results, the method of Fourier analysis (Ghedini 1982) was used.

4.1 Period analysis of absorption lines

In our previous articles (Gális et al. 1999; Friedjung et al. 2003), we performed detailed period analysis of radial velocities based on absorption-line measurements. We note that these measurements have very high accuracy, with typical errors of 0.4–0.8 km s^{-1}. The data (135 radial velocities) cover the time interval JD 244 6578.5–245 1676.9 (5098 d). We re-analysed all these data to confirm the presence of periods longer than 1000 d. We detected only two significant periods, 550.4 ± 1.4 and 355.0 ± 1.6 d, related to the orbital motion and cool-component pulsations, respectively. The power

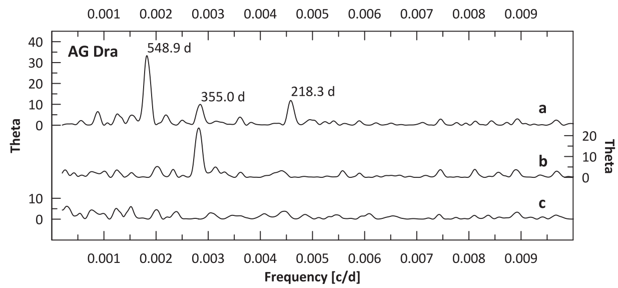

Figure 7. Power spectra of AG Dra taken from combined radial velocities based on absorption-line measurements: (a) original data and (b) data with orbital response removed, as well as (c) data with both orbital and probable pulsation response removed.

spectra taken from combined radial velocities based on absorption-line measurements are depicted in Fig. 7. We can confirm that the data did not contain variability with longer periods. Moreover, the statistical analysis suggests that data with those errors did not contain any signal other than orbital motion and postulated pulsation.

4.2 Period analysis of emission lines

We have obtained equivalent widths (EW) and absolute fluxes of spectral emission lines Hα, Hβ, He I 6678 Å, He II 4686 Å and Raman-scattered O VI 6825 Å, as well as the radial velocities of these lines (except the O VI 6825-Å line). The variability of EW of some spectral lines of AG Dra has been well-known already for many years and many authors have made a great effort to assign this variability to physical mechanisms present in the system (Kaler 1987). Up to now, this variability has not been explained without uncertainties. The variation of EW is related to the increase or decrease of the amount of observed emitted/absorbed particles. This variability would be related to the orbital motion, with a dependence on the orientation of the system towards the line of sight at a given moment (i.e. orbital phase). For more details, see Leedjärv et al. (2004).

The results of the period analysis of these data can be summarized as follows. The most significant periods detected in radial velocities, fluxes and EWs for all emission lines are close to the orbital period (511–568 d) and to the time interval between individual outbursts (366–383 d). The period related to the pulsation of the red giant (350–357 d) was marginally detected. The period analysis did not show the presence of the longer significant periods around 1160 d, as did the analysis of radial velocities based on the absorption-line measurements (see previous section). There are only a few detections of an insignificant period around 1100 d, which is the double of the orbital period. The curves of EW for particular spectral lines are depicted in Fig. 8.

5 DISCUSSION AND CONCLUSIONS

The main goal of this article was the complex and detailed period analysis of photometric and spectroscopic data of AG Dra. The photometry covers a time interval of 124 years, while the last 39 years of photometric observations are based on systematic photoelectric and CCD monitoring. Spectroscopic data were obtained from absorption- and emission-line measurements. The results of period analysis of all these data are two real periods present in this symbiotic system: 550 and 350 d, related to the orbital motion and postulated pulsation of the cool component, respectively. The orbital period is mainly manifested during the quiescent stages at

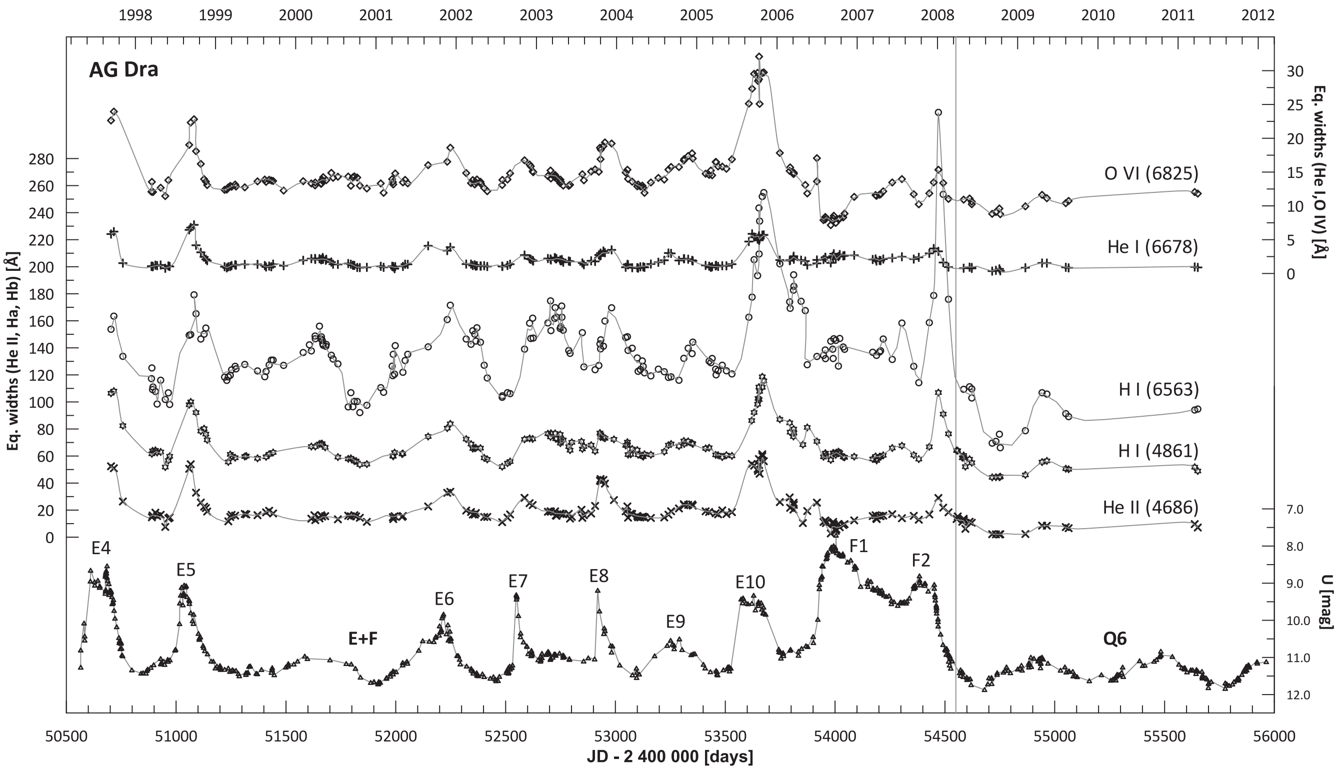

Figure 8. The curves of EW for particular spectral lines. The scales on the left and right axes are valid for EWs of He I (6678 Å) and He II (4686 Å), respectively. The thin line shows a spline fit to the data points.

shorter wavelengths (*U* band), while the pulsation period is present during quiescent as well as active stages at longer wavelengths (*B* and *V* bands). The period analysis of active stages confirmed the presence of a period of around 365 d, which is the median of the time interval between outbursts. It is worth noting that these time intervals vary from 300–400 d without an apparent long-term trend. The period analysis of the active stages also revealed longer periods (e.g. 1330, 1580, 2350, 5500 d). Our detailed analysis shows that most of these periods are more likely related to the complex morphology of the LCs during active stages than to the real variability present in this symbiotic system.

Recently, the properties of the LC of AG Dra covering a period of 120 years were studied by Formiggini & Leibowitz (2012). Their research was mainly based on photographic observations (Robinson 1969) and visual estimates from the American Association of Variable Star Observers (AAVSO) data base. It is worth noting that analysis of such data could give unreal periods (e.g. 400, 440 d periods during the Q2 and Q3 stages in our analysis). As complementary observations, Formiggini & Leibowitz (2012) used brightness measurements of AG Dra in the *U* filter from the literature. In our opinion, their proposed model is unrealistic and non-physical in some aspects. The result of the LC period analysis of these authors is the detection of the period 373.5 d, which is the mean time interval between individual outbursts in the active stages. Authors interpreted this period as the synodic rotational period of the cool giant with respect to the white dwarf (for a given point on the giant surface, the white dwarf is in upper culmination every 373.5 d). To secure such a synodic rotational period in a binary with orbital period around 550 d, the giant should rotate in retrograde fashion with a period of 1160 d. The detection of this period is also reported by these authors. We could not confirm the presence of this period in photometric as well as spectroscopic data. Moreover, such a value of the rotational period of the giant is not typical in symbiotic systems (e.g. table 2 of Formiggini & Leibowitz 2012) and the explanation for the retrograde rotation of a component in such an open system from an evolutionary point of view is unclear.

Formiggini & Leibowitz (2012) suggest that the cool giant of AG Dra has a very strong magnetic field, the axis of which is substantially (around 90 degrees) inclined relative to the rotational axis. Moreover, they suggest that the outbursts of AG Dra are the result of intensive outflow of matter from the giant towards the white dwarf. When the region of the magnetic poles of the giant gets to the tidal bulge (the strong tidal deformation of the shape of a giant in the direction towards the white dwarf), the balance is disrupted and hydrogen-rich matter is thrown into the Roche lobe of the white dwarf. This will release large amounts of gravitational energy, which becomes apparent as an outburst observed in the optical. Since the amount of matter released depends on the intensity of the magnetic field, as well as the hydrodynamic and thermodynamic properties of matter in the tidal bulge at a given time, each outburst can vary greatly. According to the opinion of specialists studying stellar magnetic activity across the Hertzsprung–Russell (HR) diagram, such very strong magnetic fields of cool giants are not known (Korhonen, private communication). We do not understand the process of balance-breaking by a tidal bulge, in view of the fact that the whole surface of the tidally deformed giant in a binary lies on the same equipotential surface. The authors explain the alternating active and quiescent stages of AG Dra by a mechanism similar to the solar dynamo (responsible for solar-cycle activity), which takes place in the outer layers of the extensive giant atmosphere. Our analysis of all 124 years of brightness history has shown that active stages have a recurrence frequency of 12–16 years, which is in good agreement with the idea of solar-like activity.

The period analysis of photoelectric and CCD observations obtained during the active stages D and E + F supported the results of our statistical analysis of the data: the LCs in the *U*, *B* and *V* bands are strongly cross-correlated during the activity of AG Dra. The obtained periods in individual bands have the same values within their errors. Global morphology of the active stages is well described by two longer periods (about 1500 and 5400 d). The obtained power spectra also contained a period of about 1100 d, but the significance of the period decreased considerably after removal of these two longer periods from the LCs of AG Dra (Fig. 4). From this fact, we suggested that the 1100-d period more likely describes only the morphology of the active stages of AG Dra, rather than a real variability presented in this system. The power spectra obtained after removal of the longer periods (about 1500 and 5400 d) showed the presence of a significant period around 375 d (Fig. 4), which is related to the outburst time distribution. As can be seen, the maximum of significance of this period has two peaks (second peak around 360 d), suggesting a possible long-term change of its value. This assumption was confirmed by detailed period analysis of individual parts of the active stage E+F. While at the beginning of E+F stage the power spectrum contains only a period of about 370 d, at the end of this active stage the power spectrum contains only a period of 360 d. The detection of a period of about 559 d (close to the value of the orbital one), which was significant during the stages of activity in the *B* and *V* bands, is also interesting.

The situation was different during the quiescent stages (Q4–Q6) and the period analysis confirmed that the LC in the *U*, *B* and *V* bands did not cross-correlate. The period analysis clearly confirmed the presence of only the orbital period (551.8 d) in the *U* band, as well as the orbital (551.7 d) and pulsation (351.2 d) periods in the *B* band (Fig. 4). The power spectrum in the *V* band (Fig. 4) is more complex: although the orbital (547.2 bottom) and pulsation (350.6 bottom) periods were detected, the resulting power spectrum also contained other significant periods (334.1, 223.0 and 136.8 d). However, this may be only an artefact caused by the fact that the amplitude of the light variations in the *V* band during quiescent stages of AG Dra is comparable to the observation errors in this band.

The interpretation of the time intervals between outbursts with a median of 365 d is more complicated. The emission spectral lines are created in the part of the symbiotic nebula originating from the stellar wind of the red giant and ionized by radiation from the white dwarf. The stellar wind should be modulated by the giant pulsations. This modulation produces a Doppler shift of the spectral lines, which is manifested in the radial velocities. Simultaneously, the number of emitted/absorbed particles changes what can be detected as regards EW variability. Skopal et al. (2009) proposed another model, where the hydrogen lines are created in the wind of the hot component. In this case, the main feature is the mutual interactions of the wind on both components, but in this model it is impossible to explain all observed effects.

Leibowitz & Formiggini (2006, 2008, 2011, 2013) and Formiggini & Leibowitz (2006, 2012) have found similar patterns in long-term LCs of six symbiotic stars, including AG Dra. The common feature of all these stars is the start of active periods at about 4650–7550 d intervals (12–20 years). This implies a common physical mechanism: for example, solar-like magnetic cycles might be responsible for the activity of these symbiotic stars. However, there is no direct evidence of the presence of a reasonably strong magnetic field in the red giants in these systems. Complex morphology of the LCs may introduce artefacts in their period analysis. We suggest that a careful case-by-case analysis of the LCs, together with spectroscopic data and, when possible, observations in the wide wavelength range from X-rays to radio, would give the first clues to understanding the physical mechanism of the outburst behaviour of symbiotic stars. One should not pay too much attention to the possible artificial periods obtained from the analysis of complicated LCs. Outburst phenomena are not necessarily strictly periodic, as we have seen in the case of AG Dra. Let us recall that all the active periods of AG Dra have been different from each other (Figs 1 and 3).

One promising explanation of at least some individual outbursts of AG Dra might be the combination nova model proposed for Z And by Sokoloski et al. (2006). In this model, when the accretion rate on to the white dwarf exceeds some critical value, thermonuclear reactions are ignited and the luminosity of the hot component increases significantly. One of the next tasks would be to study whether hot and/or cool outbursts of AG Dra will fit into such a picture. We will present a detailed analysis of the spectroscopic data of AG Dra in our forthcoming article.

ACKNOWLEDGEMENTS

We dedicate this article to the memory of our friend Michael Friedjung (1940–2011), long-term collaborator on AG Dra and other symbiotic stars. Michael passed away suddenly, but we are sure that his wonderful spirit remains with us.

This study was supported by the Slovak Academy of Sciences VEGA Grant No. 2/0038/13, by the realisation of the Project ITMS No. 26220120029 and by the Estonian Ministry of Education and Research target financed research topic SF0060030s08. This article was also supported by the realisation of the Project ITMS No. 26220120029, based on the supporting operational research and development program financed from European Regional Development Fund.

REFERENCES

Andronov I. L., 1994, Odessa Astronomical Publications, 7, 49
Bastian U., 1998, A&A, 329, L61
Belyakina T. S., 1965, Izv. Krymskoj Astrofiz. Obs., 33, 226
Belyakina T. S., 1969, Izv. Krymskoj Astrofiz. Obs., 40, 39
Breus V. V., 2003, Odessa Astronomical Publications, 16, 24
Contini M., Angeloni R., 2011, New Astron., 16, 199
Fekel F. C., Hinckle K. H., Joyce R. R., Skrutskie M. F., 2000, AJ, 120, 3255
Formiggini L., Leibowitz E. M., 2006, MNRAS, 372, 1325
Formiggini L., Leibowitz E. M., 2012, MNRAS, 422, 2648
Friedjung M., Hric L., Petrík K., Gális R., 1998, A&A, 335, 545
Friedjung M., Gális R., Hric L., Petrík K., 2003, A&A, 400, 595
Gális R., Hric L., Friedjung M., Petrík K., 1999, A&A, 348, 533
Garcia M. R., 1986, AJ, 91, 1400
Ghedini S., 1982, Software for Photometric Astronomy. Willmann-Bell, Richmond, VA
González-Riestra R., Viotti R., Iijima T., Greiner J., 1999, A&A, 347, 478
Hric L., Skopal A., 1989, Inf. Bull. Variable Stars, 3364, 1
Hric L. et al., 1996, Contrib. Astron. Obs. Skalnaté Pleso, 26, 46
Huang C. C., Friedjung M., Zhou Z. X., 1994, A&AS, 106, 413
Kaler J. B., 1987, AJ, 94, 437
Kenyon S. J., Fernandez-Castro T., 1987, AJ, 93, 938
Leedjärv L., Burmeister M., Mikolajewski M., Puss A., Annuk K., Galan C., 2004, A&A, 415, 273
Leedjärv L., Burmeister M., 2012, Baltic Astronomy, 21, 131
Leibowitz E. M., Formiggini L., 2006, MNRAS, 366, 675
Leibowitz E. M., Formiggini L., 2008, MNRAS, 385, 445
Leibowitz E. M., Formiggini L., 2011, MNRAS, 414, 2406

Leibowitz E. M., Formiggini L., 2013, AJ, 146, L117
Luthardt L., 1983, Mitteilungen über Veränderliche Sterne, 9, 129
Meinunger L., 1979, Inf. Bull. Variable Stars, 1611, 1
Mikolajewska J., Kenyon S. J., Mikolajewski M., Garcia M. R., Polidan R. S., 1995, AJ, 109, 1289
Munari U. et al., 2009, PASP 121, 1070
Press W. H., Teukolsky S. A., Vetterling W. T., Flannery B. P., 1992, Numerical Recipes: The Art of Scientific Computing, Second Edition. Cambridge Univ. Press, New York
Robinson L., 1969, Peremennye Zvezdy, 16, 507
Schwarzenberg-Czerny A., 1991, MNRAS, 253, 198
Shore S. N., Genovali K., Wahlgren G. M., 2010, Baltic Astronomy, 21, 139
Shore S. N. et al., 2012, A&A, 510, 70
Skopal A., 1998, Contrib. Astron. Obs. Skalnaté Pleso, 28, 87
Skopal A., Chochol D., 1994, Inf. Bull. Variable Stars, 4080, 1
Skopal A. et al., 1995, Contrib. Astron. Obs. Skalnaté Pleso, 25, 53
Skopal A., Vaňko M., Pribulla T., Wolf M., Semkov E., Jones A., 2002, Contrib. Astron. Obs. Skalnaté Pleso, 32, 62
Skopal A., Vaňko M., Pribulla T., Velič Z., Semkov E., Jones A., 2004, Contrib. Astron. Obs. Skal. Pleso, 34, 45
Skopal A., Sekeráš M., González-Riestra R., Viotti R. F., 2009, A&A, 507, 1531
Skopal A., Shugarov S., Vaňko M., Dubovský P., Peneva S. P., Semkov E., Wolf M., 2012, Astronomische Nachrichten, 333, 242
Smith V. V., Cunha K., Jorissen A., Boffin H. M. J., 1996, A&A, 315, 179
Sokoloski J. L., Kenyon S. J., Espey B. R., 2006, ApJ, 636, 1002
Splittgerber E., 1974, Mitteilungen über Veränderliche Sterne, 6, 193
Zamanov R. K., Bode M. F., Melo C. H. F., Bachev R., Gomboc A., Porter J., Pritchard J., 2007, MNRAS, 380, 1053

This paper has been typeset from a TEX/LATEX file prepared by the author.